

1 Perceptual distinctiveness predicts the lexical distribution 2 of vowels: Evidence from 30 languages

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8 Abstract

9 Perceptual distinctiveness is known to influence vowel inventories, but does it also
10 shape the lexical distribution of vowels? This paper investigates whether the frequency
11 of vowel qualities in lexicons can be predicted by perceptual factors, i.e., whether more
12 distinct vowels are used more frequently in words. Using phonetic data from a geograph-
13 ically and genetically diverse sample of 30 languages from DoReCo, we compare three
14 models: a baseline model assuming equiprobability, and two phonetically grounded
15 models based on Dispersion Theory and Focalization Theory, respectively. Our results
16 show that Dispersion Theory best predicts vowel counts, providing evidence that per-
17 ceptual distinctiveness extends to lexical distribution. Consistent with prior research on
18 vowel inventories, we find that $F1$ must outweigh $F2'$ in perceptual weighting. We dis-
19 cuss the implications of these findings for phonetically based theories of vowel systems
20 and models of sound change.

21 **Keywords:** dispersion; focalization; lexicon; frequency; typology

22 **Word count:** 12,312

1. Introduction

Perceptual distinctiveness has been a central theme in theories on sound inventories since Liljencrants and Lindblom (1972). In their seminal paper, Liljencrants and Lindblom proposed that vowel inventories are shaped by the principle of maximum perceptual contrast: vowels must be maximally distinct in order to support phonological contrasts—a theory known as Dispersion Theory (e.g., Becker-Kristal 2010; Flemming 2017; Schwartz et al. 1997). The theory successfully accounts for some crosslinguistic trends in the structure of vowel inventories (Becker-Kristal, 2010; Gordon, 2016), explaining for instance why corner vowels [a, i, u], which are the three most perceptually distinct vowel qualities that can be produced by the human vocal tract, are also the most frequently attested vowels across sound inventories of the world’s languages (see the PHOIBLE database; Moran and McCloy 2019).

Beyond shaping sound inventories, perceptual distinctiveness affects other domains, such as phonotactics and phonological alternations (e.g., Flemming 2004; Padgett 2003). This paper tests whether perceptual distinctiveness also impacts the lexicon, i.e., whether sounds which are perceptually more distinct are also used more frequently across words within languages. This hypothesis is supported by the observation that language-internal sound frequencies mirror cross-linguistic patterns (Gordon, 2016), as well as by existing theories of sound change, whether speaker- (Lindblom, 1990) or listener-based (Ohala, 1981). However, this has not been thoroughly tested using the same mathematical models proposed for sound inventories.

To address this gap, we focus on vowels, where evidence for perceptual distinctiveness at the level of inventories is clearest (on consonant inventories, see Flemming 2004; Lindblom and Maddieson 1988 among others). We adapt the phonetically based model of vowel inventories proposed by Schwartz et al. (1997) and use it to predict vowel lexical frequencies from acoustic data. As a testing ground, we use a 30-language sample from the multilingual speech corpus DoReCo (Seifart et al., 2024). We implement the analysis in a probabilistic constraint-based framework, using Schwartz et al. (1997)’s phonetically based constraints to model perceptual distinctiveness in the $F1:F2'$ space and maximum entropy grammars with systemic constraints to model counts (Goldwater & Johnson, 2003; Hayes & Wilson, 2008; Storne, 2026). We evaluate the effect of perceptual distinctiveness on the lexical frequency

of vowels by comparing three models: (i) a null model that predicts equiprobability, (ii) a phonetically based model that favors perceptually distinct vowels, in line with Dispersion Theory, and (iii) another phonetically based model that favors focal vowels, in line with Quantal/Focalization Theory, an alternative theory to explain crosslinguistic tendencies in vowel inventories (Schwartz et al., 1997; Stevens, 1989). We further test two variants of each model: one with universal weighting favoring lower vowel formants ($F1$ in Dispersion Theory and $F1$ - $F2$ convergence in Focalization Theory), and another with language-specific weighting. The results show that the dispersion model with language-specific weights provides the best fit to the data, confirming that perceptual distinctiveness plays a role within languages—but likely in a language-specific manner.

Section 2 provides background on perceptual distinctiveness in vowel inventories and introduces the hypothesis that it may also shape lexicons. Section 3 describes the corpus and methods used to test this hypothesis. Section 4 presents the lexical frequency and phonetic realization data for vowels, followed by the model comparison. Finally, Section 5 explores the implications of the results for phonetically based theories of vowel inventories and sound change. It also highlights the limitations of this work and outlines future directions.

2. Background on perceptual distinctiveness: from inventories to lexicons

2.1. Perceptual distinctiveness in vowel inventories

Languages use different sounds to build words. For instance, French features the vowel [y] in its vowel inventory, as in the word *mur* ‘wall’ [myʁ], whereas English does not. Conversely, [ʌ] is present in English, as in the word *cut* [kʌt], whereas it is absent in French. Yet, despite this variety, certain types of sounds are consistently favored across sound inventories (Gordon, 2016). For example, vowels [a, i, u] are present in the vast majority of languages (over 80% of languages in the PHOIBLE database) whereas [y, ʊ, ʌ] are far less common (present in fewer than 10% of languages in PHOIBLE; Moran and McCloy 2019).

A number of works have suggested that these crosslinguistic tendencies in sound inventories are shaped by perceptual and articulatory factors (e.g., among many others Flemming 2002; Liljencrants and Lindblom 1972; Lindblom 1986; Lindblom and Maddieson 1988; Martinet 1955/1970; Stevens 1989). Among these factors, the principle of maximal perceptual contrast—the preference for acoustically distinct sounds, making them easier for

listeners to differentiate in perception—has long been recognized as key in shaping vowel inventories (e.g., Martinet 1960/1980, p. 200; see Becker-Kristal 2010 for a comprehensive review). For instance, the high crosslinguistic frequency of [a, i, u] has been explained by the fact that these sounds are maximally dispersed in the auditory space of possible vowel qualities and are therefore the most perceptual distinct vowels that can be produced by the human vocal tract. By contrast, [ʌ, y, ʊ] are less frequently used in vowel inventories because they occupy more central positions in the auditory space and are therefore typically less distinct from other vowels present in languages.

This intuitive notion of perceptual distinctiveness has been made precise and various explicit models have been proposed to derive the typology of vowel inventories (Cotterell & Eisner, 2018; De Boer, 2001; Flemming, 2002, 2017; Liljencrants & Lindblom, 1972; Sanders & Padgett, 2008; Schwartz et al., 1997). Among these, Liljencrants and Lindblom (1972)’s Dispersion Theory and Schwartz et al. (1997)’s Dispersion-Focalization Theory have been particularly influential.

Liljencrants and Lindblom (1972)’s Dispersion Theory derives predictions about vowel inventories by maximizing the aggregate Euclidean distance between all vowels in a two-dimensional auditory space, corresponding to the vowel first formant ($F1$) and a combination of the second and third formants ($F2'$). The use of the lower formants ($F1$, $F2$, $F3$) as predictors is motivated by the role these play in the perception of vowel-quality contrasts (e.g., Delattre et al. 1952). The model achieves a very good fit for small vowel inventories. Inventories [i, a, u] and [i, ɛ, a, u] are derived as the optimal three- and four-vowel inventories, aligning closely with typological observations. However, the model’s predictive accuracy diminishes for larger inventories, as the model tends to generate fewer vowels on the height dimension ($F1$) and more vowels on the front-back dimension ($F2'$) than what is typically observed (see also Lindblom 1986, p. 21; Gordon 2016, pp. 58–60).

Schwartz et al. (1997) proposed a revised theory, called the Dispersion-Focalization Theory, to overcome some of the shortcomings of Dispersion Theory (see Gordon 2016, pp. 60–62 for a review). Besides the addition of a new factor, focalization, which favors vowels with formant convergence (see Section 2.3 for more details), the main difference with Dispersion Theory is the asymmetric weighting of formant frequencies in the calculation of inter-vowel Euclidean distances. This allows dispersion along the height dimension ($F1$) to matter more

than dispersion along the front-back dimension ($F2'$). Schwartz et al. (1997) demonstrate that, under the assumption that $F1$ is more weighted in the dispersion calculation, the theory accurately accounts for the structure of vowel inventories, at least for inventory sizes ranging from three to seven vowels. The importance of $F1$ for vowel contrasts has been interpreted as reflecting its higher intensity (Lindblom, 1986, p. 28) and its greater resistance to noise (Schwartz et al., 1997).

The Dispersion-Focalization Theory represents a significant advancement over the original Dispersion Theory but retains some of its limitations. For instance, the theory fails to derive schwa systems, a gap that may stem from its neglect of articulatory effort minimization, a factor known to influence the distribution of central vowels like schwa or to drive vowel reduction (e.g., Schwartz et al. 1997, p. 260, Flemming 2004, p. 246, Gordon 2016, pp. 61–62). Additionally, this theory does not explain the symmetry frequently observed in vowel inventories, where vowels tend to be evenly distributed along both the front-back and height dimensions (Schwartz et al., 1997, pp. 280–281). Also, the model is limited to predicting vowel qualities; yet larger vowel inventories often make use of other phonetic dimensions, such as duration, nasalization, etc. These limitations suggest that richer models are necessary to fully capture the typological diversity of vowel systems. Despite their limitations, these studies by Liljencrants and Lindblom (1972) and Schwartz et al. (1997) have contributed to establishing perceptual distinctiveness as a crucial factor to explain vowel inventories, and will serve as the basis for this work.

2.2. Beyond inventories: perceptual distinctiveness in the lexicon

Perceptual distinctiveness effects have also been argued to extend beyond inventories, influencing phonotactics and phonological alternations (e.g., Flemming 2002, 2004, 2017; Padgett 2003; Stanton 2017; Storme 2019). For instance, Flemming (2004) argues that the neutralization of vowel height contrasts in unstressed syllables—as observed in languages like Italian—arises from reduced perceptual distinctiveness caused by duration-induced target undershoot. To maintain sufficient dispersion, the size of vowel inventories is reduced in unstressed contexts, resulting in neutralization patterns and stress-conditioned alternations.

Another pattern driven by perceptual distinctiveness is contrast enhancement. For instance, Storme (2019) argues that patterns of vowel tensing enhance perceptual distinctive-

ness along the front-back dimension (F2), as in Southern French, where underlying /ɛ/ is realized as [e] and underlying /ɔ/ as [o] in open syllables (e.g., *fêt-ard* /fɛt-aʁ/ ‘party goer’ realized as [fɛtaʁ]). This reflects a strategy to increase distinctiveness, as the [e]-[o] contrast is more distinct along F2 than the [ɛ]-[ɔ] contrast.

This paper explores another potential domain where perceptual distinctiveness may play a role: the lexicon. The investigation is motivated by a striking parallel—asymmetries in usage frequency observed not only across vowel inventories in the world’s languages but also within the lexical distributions of individual languages (e.g., Everett 2018; Gordon 2016). Sounds that occur frequently in sound inventories (e.g., corner vowels [a, i, u]; consonants [n, k, t]) also tend to be used frequently in the lexicons of languages, whereas sounds that are crosslinguistically rare (e.g., central vowels [ʌ, y, ʊ]; consonants [f, ɲ, ɖʒ]) tend to be used unfrequently in languages that have them (Gordon, 2016, pp. 71–76).

This correlation suggests that articulatory and perceptual factors that shape sound inventories also shape lexicons. Among these factors, perceptual distinctiveness is likely to play an important role. For instance, the fact that corner vowels [a, i, u] are frequently used in the lexicons of languages that have these sounds makes sense if lexical distributions of sounds are optimized for perceptual distinctiveness.

The existence of an effect of perceptual distinctiveness on the lexicon would come as no surprise, given existing causal theories of how this effect comes about in sound inventories. In listener-based models (e.g., Blevins 2004; Ohala 1981), the structure of sound systems is influenced by sound change across generations. Perceptual distinctiveness in sound inventories arises from the cumulative effect of misperceptions over time: sounds involved in poorly distinct contrasts are more likely to be misperceived by listeners, leading to a decline in their lexical frequency. Over time, these sounds may be lost altogether from the lexicons of languages, leading to a decline in their frequency across the sound inventories of the world’s languages. In other words, these models ultimately derive asymmetries in sound inventories from asymmetries in lexicons, and therefore predict a correlation between sound frequency within and across languages.

This correlation is also consistent with predictions from speaker-based models of dispersion. Contrasting with listener-based approaches, these models assume that speakers actively (though unconsciously) adjust their speech production to enhance perceptual clar-

ity. This assumption is motivated by phenomena such as the so-called Lombard effect, where speakers adjust their production in noisy environments (see Elie et al. 2024). In speaker-based models of sound inventories, perceptual distinctiveness of phonological contrasts arises through two primary mechanisms: a phonological production bias that favors perceptually distinct sounds (e.g., Flemming 2017; Lindblom 1990 or Boersma and Hamann 2008 for a related variant) or a lexical production bias that discourages ambiguous word realizations (e.g., Wedel 2012; see Flemming 2017 for a discussion of these models).

In the phonological-bias approach, production biases are proposed to extend from individual sounds to lexical items through phonologically driven lexical competition (Martin, 2007). Specifically, when multiple lexical candidates could express a given concept, those that align more closely with phonological biases (such as featuring easily producible or perceptually distinct sounds) are more likely to be selected. This mechanism predicts that perceptual distinctiveness should influence not only the structure of sound inventories but also the frequency distribution of sounds within lexicons.

This effect is also predicted by the lexical-bias approach proposed by Wedel (2012), though with reversed causality: rather than phonological biases shaping lexical selection, this model posits that a preference for lexically unambiguous words drives the selection of more acoustically distinct sounds within those words. Over time, this lexical bias feeds back into the sound system, gradually shaping the structure of the language's sound inventory.

To sum up, theories that have been proposed to explain how perceptual distinctiveness shapes sound inventories also predict that it should shape lexicons.

2.3. The need for an explicit model

However, this prediction has not been tested thoroughly and quantitatively through explicit modeling studies of the kinds that have been proposed for vowel inventories. Although indicative, the study by Gordon (2016) showing that corner vowels [a, i, u] are more frequent than central vowels (e.g., [ʌ, y, ʊ]) in the lexicons of languages is not sufficient to establish the role of perceptual distinctiveness in the lexicon. The first obvious reason is that distinctiveness is a system-dependent property. A corner vowel like [i] could be suboptimal from a dispersion perspective in a language where it contrasts with [ɪ]. The effect of dispersion on the lexical frequency of vowels must be established while taking into account specific

systems of contrasts in languages.

Another reason is that perceptual distinctiveness must be established using phonetic data. It cannot be established solely based on IPA labels, as these do not provide reliable information about phonetic realization. For instance, IPA labels do not necessarily reflect the same vowel quality across languages (e.g., Bradlow 1995). Also, transcription choices are partly motivated by language-internal considerations. For instance, if a language does not make a phonemic distinction between [i] and [ɪ], it is more common to transcribe the vowel as [i], regardless of its actual phonetic realization (see Cotterell and Eisner 2018, in the context of testing Dispersion Theory). The phonetic label [i] can then not be considered as always reflecting the same vowel quality across languages. The effect of perceptual distinctiveness on the lexical frequency of vowels must be assessed by taking into account specific patterns of phonetic realization in languages.

Finally, other factors than perceptual distinctiveness/dispersion could explain why [a, i, u] are frequent within languages. One obvious factor to take into account is focalization, as it has been proposed to also favor corner vowels (Johnson 2012, pp. 141–142; Schwartz et al. 1997; Stevens 1989). Distinctiveness/dispersion relies on the relative position of vowels in perceptual space. On the other hand, focalization is grounded in an intrinsic notion of perceptual salience. Focal vowels are characterized by formant convergence, where two or more formant frequencies (e.g., F_1 and F_2 , F_2 and F_3 , or F_3 and F_4) are similar, creating a strong spectral peak. This convergence enhances their perceptual salience. For instance, in an experiment manipulating the distance between F_2 and F_3 and the distance between F_3 and F_4 , Schwartz and Escudier (1989) found that vowels with greatest formant convergence were easier to memorize by listeners than vowels with less convergence.

Focalization could favor corner vowels [a, i, u] because these vowels are characterized by formant convergence: F_1 and F_2 convergence for [a] (due to its high F_1 target) and [u] (due to its low F_2 target) and F_3 and F_4 convergence for [i]. Focalization has also been used to explain why some relatively poor vowel contrasts such as [y]-[i] are relatively frequent in languages. This can be explained because [y] exhibits high focalization due to the proximity of its F_2 and F_3 frequencies (Stevens, 1989, p. 33). Focalization is related to the Quantal Theory of Stevens (1989), a theory which predicts a preference for articulations that have stable acoustic consequences. The two notions (focal and quantal vowels) are

connected because quantal regions in articulation (regions where small articulatory changes have little acoustic effects) tend to coincide with focal regions in acoustics (regions with convergence of formants). Focal vowels could therefore also have an advantage in that they are articulatorily more stable. The key implication is that any effect of dispersion on the lexical frequency of vowels should be compared to the effect of focalization.

3. Method

3.1. Language sample

We used data from 30 languages from the Language Documentation Reference Corpus (DoReCo; Seifart et al. 2024). DoReCo is a multilingual speech corpus including data from a world-wide sample of 53 languages. It includes corpora that originated in fieldwork-based documentations of small and endangered languages. We focus on a sample of 30 languages, listed in Table 4 in the appendix, along with some descriptive information (language family, area, etc.). This sample was selected based on data availability and corpus size, prioritizing languages directly available on the DoReCo website and with larger corpora. This sample remains genetically and geographically diverse, as shown in Table 4 and in the map in Figure 1.

For this analysis, we rely on phonetic transcriptions and word segmentation provided by DoReCo (tiers *ph* and *wd*, respectively). Although these may not perfectly align with contrastive phonological units (phonemes) or lexical units (morphemes) due to phonological processes and complex morphology, obtaining such units is challenging. Morphological information is not always available in DoReCo, and phonemic analysis would require additional data (e.g., stress patterns for vowels) that are not systematically provided. We address these limitations further in the discussion (Section 5).

3.2. Vowel inventories

The analysis concentrates on inventories of oral monophthongal vowels, in line with previous works testing the effects of dispersion and focalization against large typological databases (Becker-Kristal, 2010; Cotterell & Eisner, 2018; Liljencrants & Lindblom, 1972; Schwartz et al., 1997). Concretely, this means that consonants as well as vowels involving dynamic cues



Figure 1. Map of the languages

or other dimensions beyond vowel quality were excluded. This includes among others diphthongs (found in Arapaho and Mojeño Trinitario), nasal vowels (found in Anal and Swiss French), or pharyngealized vowels (found in Sanzhi Dargwa).

In line with previous work on dispersion and focalization, we also ignore vowel duration. However, instead of conflating long and short vowels together (e.g., by treating [i] and [i:] as representing the same vowel quality [i]), as done in some typological work (Gordon, 2016, p. 50), we exclude long vowels from the analysis and focus entirely on short vowels. This choice is motivated by the existence of asymmetries in the structures of inventories of long and short vowels. Failing to control for these asymmetries may make it difficult to evaluate the effect of perceptual distinctiveness on lexical frequency. For instance, in our language sample, Evenki does not have the same number of short and long vowels: it has 5 short vowels (a, ə, i, o, u) but 6 long vowels (a:, e:, ə:, i:, o:, u:) (Paschen et al., 2022). Long [e:] lacks a short counterpart. This asymmetry could lead to a low lexical frequency for the vowel quality [e] independently of the effect of dispersion or focalization. Additionally, all languages in the sample have short vowels but not all of them have long vowels.

Vowels that are only attested in loanwords (according to the DoReCo documentation) were also excluded from the analysis, due to evidence that native words and loanwords be-

long to different phonological subsystems (Itô & Mester, 2017; Kawahara & Kumagai, 2024). Also, sounds that are only attested in loanwords could have a lower lexical frequency due to their loanword status independently of the effect of acoustic dispersion and focalization. Following this reasoning, the vowel [o] was excluded from the vowel inventory of Cashin-
auha, as it only occurs in Spanish loanwords.

The resulting vowel inventories are shown in Table 1. Inventory size ranges from 3 to 12 vowels, with five-vowel inventories being the most common type, as found in the UCLA Phonological Segment Inventory Database (UPSID; Gordon 2016, p. 50). Figure 2 shows the number of languages possessing the 20 oral vowel qualities attested in the sample. The three most common vowels are the three corner vowels [a, i, u], as found in larger typological databases such as Phoible (Moran & McCloy, 2019).

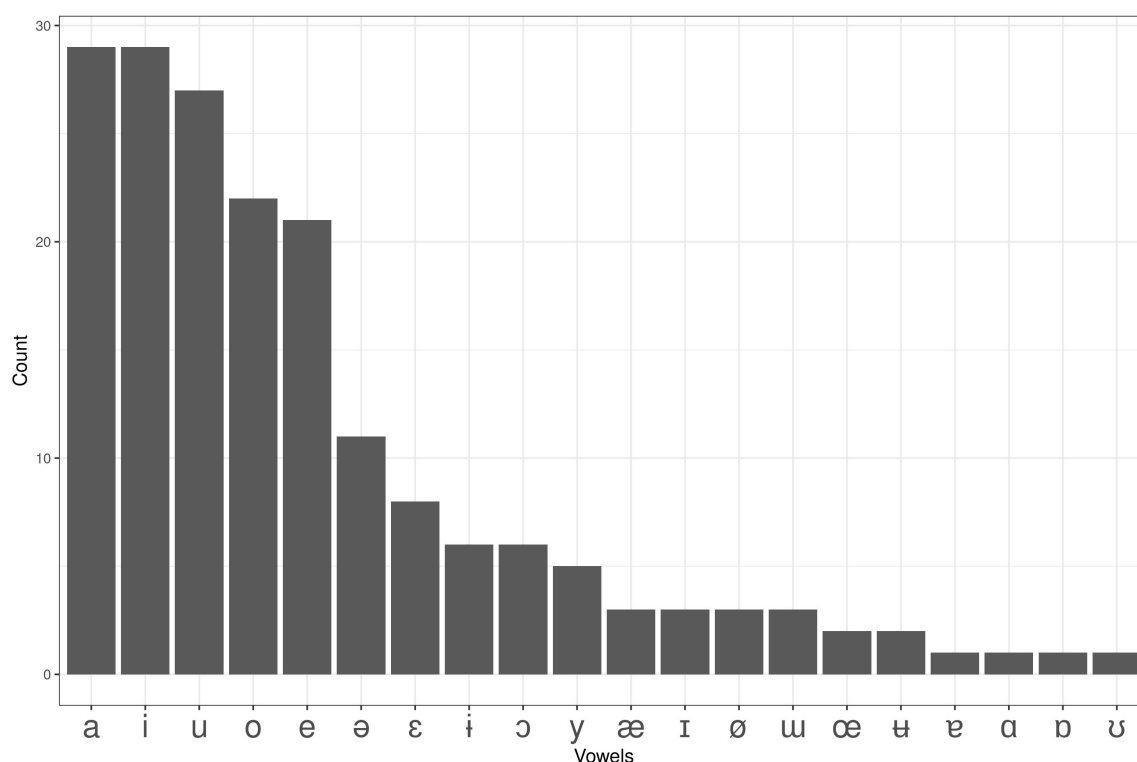


Figure 2. Number of languages in the sample possessing the 20 short oral vowels

3.3. Measures

The goal of this paper is to evaluate the effect of perceptual properties of vowels (dispersion and focalization) on their frequency of use within languages. Therefore, both phonetic measures and usage frequency metrics are necessary.

Language	Glottocode	N	Inventory
Anal	anal1239	7	a, e, ə, i, o, u, ʊ
Arapaho	arap1274	4	ɛ, ɪ, ɔ, ʊ
Bainounk Gubëeher	bain1259	10	a, e, ə, ɛ, i, ɪ, o, ɔ, u, ʊ
Beja	beja1238	5	a, e, i, o, u
Bora	bora1263	6	a, ɛ, i, i, o, ʊ
Cashinahua	cash1254	4	a, ə, i, u
Dalabon	ngal1292	6	a, e, i, i, o, u
Dolgan	dolg1241	8	a, e, i, i, o, ø, u, y
Evenki	even1259	5	a, ə, i, o, u
French (Swiss)	stan1290	12	a, ɑ, e, ə, ɛ, i, o, ø, œ, ɔ, u, y
Gorwaa	goro1270	5	a, e, i, o, u
Jejuan	jeju1234	9	a, ɒ, e, ə, ɛ, i, i, o, u
Kakabe	kaka1265	7	a, e, ɛ, i, o, ɔ, u
Kamas	kama1351	9	a, ɐ, e, ə, i, ɪ, i, o, u
Komnzo	komn1238	9	a, æ, e, ə, i, o, ø, u, y
Light Warlpiri	ligh1234	3	a, i, u
Mojeño Trinitario	trin1278	5	a, e, i, o, u
Nafsan (South Efate)	sout2856	5	a, e, i, o, u
Nisvai	nisv1234	6	a, ə, ɛ, i, ɔ, u
Northern Alta	nort2875	5	a, ə, i, o, u
Resígaro	resi1247	5	a, e, i, o, ʊ
Ruuli	ruul1235	5	a, e, i, o, u
Sanzhi Dargwa	sanz1248	4	a, e, i, u
Savosavo	savo1255	5	a, e, i, o, u
Tabasaran	taba1259	5	a, e, i, u, y
Texistepec Popoluca	texi1237	6	a, ə, ɛ, i, ɔ, u
Urum	urum1249	9	a, æ, e, i, o, œ, u, ʊ, y
Warlpiri	warl1254	3	a, i, u
Yali (Apahapsili)	apah1238	5	a, e, i, o, u
Yurakaré	yura1255	7	a, æ, e, i, i, o, u

Table 1. Vowel inventories of short oral monophthongs in the 30 languages analyzed

Two usage frequency metrics were collected using the transcriptions of the corpora: token and type counts. Token count corresponds to the number of times a vowel is attested in the corpus. Type count corresponds to the number of times a vowel is attested in the lexicon of the language. The lexicon of the language itself is extracted from the corpus by retaining only one occurrence of each word. In this work, we focus on type counts because we are examining the lexical distribution of vowels, but we will also mention token counts (see Section 4.1).

Following earlier works on Dispersion Theory (e.g., Liljencrants and Lindblom 1972; Schwartz et al. 1997), we used acoustic rather than perceptual measures to model dispersion and focalization. Specifically, the analysis focuses on the first four formants ($F1$, $F2$, $F3$,

F4), following Schwartz et al. (1997). Formant values were extracted using Fast Track (Barreda, 2021). Fast Track is a formant tracker implemented in Praat (Boersma & Weenink, 2025) that attempts to automatically select the best analysis from a set of candidates. For more accurate results, sound files with multiple speakers were excluded, meaning that the corpus used for acoustic measurements may be smaller than the corpus used to obtain usage frequency metrics, depending on whether the corpus contains sound files with multiple speakers. Subsequently, Fast Track was run for each speaker separately (see <https://github.com/santiagobarreda/FastTrack/wiki/How-to-analyze-a-folder>). We adhered to the plugin author's recommendations for formant frequency ranges, setting them to 4500–6500 Hz for male speakers and 5000–7000 Hz for female speakers (<https://github.com/santiagobarreda/FastTrack/wiki/getting-a-good-analysis>). Formants for each vowel were measured across five different time slices through the whole vowel duration. For the analysis, we focus on the median formant values corresponding to the middle portion of the vowel (40%-60%).

For outlier removal, we used the modified Mahalanobis distance method proposed by Stanley (2020), implemented as an R function (https://rdr.io/github/JoeyStanley/joe-yr/man/find_outliers.html). Before applying the outlier removal function, we grouped the data by language, speaker and vowel. When performing outlier removal, we required 90% of the data to be kept (and therefore 10% to be filtered out). For the analysis, the Bark transformation was used to convert the acoustic frequency scale (Hz) into an auditory frequency scale that more closely reflects how the human ear processes sound. Following Schwartz et al. (1997), we used the Bark transformation proposed by Schroeder et al. (1979). The resulting values were subsequently averaged for each language, providing an estimate of vowel phonetic realization in that language. These by-language averages are the phonetic predictors that were used to predict vowel counts in each language.

3.4. Models

We compare three theories of vowel counts in the lexicon:¹

- a baseline model, without any bias towards dispersion or focalization

¹We also considered a fourth model combining dispersion and focalization and corresponding to the Dispersion-Focalization Theory proposed by Schwartz et al. (1997). However, this model failed to converge. Further details are provided in the supplementary materials. Since our primary objective was to assess the role of perceptual distinctiveness, rather than to determine whether additional factors such as focalization are required, we do not consider the non-inclusion of the Dispersion-Focalization Theory in the paper to be a major limitation.

- the dispersion model, with a constraint favoring perceptually distinct vowels (Liljencrants & Lindblom, 1972)
- the focalization model, with a constraint favoring focal vowels (i.e. with formants in close proximity; Stevens 1989)

Throughout, we closely follow the implementation of dispersion and focalization constraints proposed by Schwartz et al. (1997) to account for crosslinguistic trends in vowel inventories. We use the maximum entropy framework (MaxEnt; Hayes and Wilson 2008) for constraint-based grammars to link phonetic constraints to vowel probabilities. MaxEnt is adopted here as the probabilistic grammar framework because prior research has shown it to be effective in modeling phonological variation (Flemming, 2021). Also, MaxEnt is based on the exponential function (see Section 3.4.2) and the exponential distribution has been argued to provide ‘a plausible model for phonological segment frequencies’ (Macklin-Cordes & Round, 2020, p. 8). Although MaxEnt is often interpreted as a cognitive model of speaker production in theoretical phonology, we do not adopt this interpretation here. This is because potential effects of perceptual distinctiveness on the lexicon can also be explained diachronically (see Section 2.2). Instead, we use MaxEnt as a tool to model lexical vowel frequency based on phonetic factors, following a similar approach to Liljencrants and Lindblom (1972) and Schwartz et al. (1997) and remaining agnostic as to whether these effects are better accounted for as listener- or speaker-based.

3.4.1. Baseline model

The baseline model predicts that all vowels are equiprobable within each language. It is the simplest model and does not include any parameter. Although this model is not a plausible representation of frequency distributions of sounds (Gordon, 2016; Macklin-Cordes & Round, 2020; Sigurd, 1968), it still serves as a convenient baseline for comparison. Moreover, it may even outperform alternative models. For instance, if competing models predict distributions that are completely misaligned with observed data—such as ranking the most frequent sounds as the least frequent, and vice versa—this model could still provide a more accurate reference point.

3.4.2. Dispersion model

The dispersion model includes a constraint favoring vowels that are more distinct. The perception of vowel-quality contrasts is assumed to occur in the Bark-transformed $F1 \times F2'$ space, where $F2'$ is the ‘perceptual second formant’. $F2'$ is recognized as a reasonable approximation of $F2$ and higher formants in determining vowel quality. $F2'$ is set at $F2$, the center of gravity of $F2$ and $F3$, or the center of gravity of $F3$ and $F4$, depending on the pattern of distances between $F2$, $F3$, and $F4$. The reader is referred to Schwartz et al. (1997, p. 263) for the specific algorithm used.

The perceptual distinctiveness of the contrast between two vowels, V_i and V_j , is quantified as the Euclidean distance between their positions in the Bark-transformed $F1 \times F2'$ space, denoted as $d_{i,j}$, as shown in 1.

$$(1) \quad d_{i,j} = \sqrt{(F1_i - F1_j)^2 + \lambda^2 (F2'_i - F2'_j)^2}$$

Following Schwartz et al. (1997), we introduce a free parameter, λ , to balance the respective contributions of $F1$ and $F2'$ when calculating the distance between two vowels. This asymmetric weighting of formant frequencies in dispersion is motivated by the empirical observation that vowel contrasts are more robustly expressed along the height dimension ($F1$) than along the front-back dimension ($F2'$) in vowel inventories—a pattern that is notoriously not captured by Liljencrants and Lindblom’s original model (see Section 2.1). In order to give more weight to $F1$ in the dispersion calculation, we constrain λ to the $[0,1]$ interval, using a Beta prior.

We further compare two variants of this model: one where λ is constrained to be identical across languages and one where λ may be language-specific. This latter possibility provides extra-flexibility in the modeling, and implements the observation that cue weighting might vary by language (e.g., Schertz et al. 2020).

The perceptual cost of a pair of vowels V_i and V_j , $disp_{i,j}$, is calculated as the inverse-square of its perceptual distance $d_{i,j}$, following Liljencrants and Lindblom (1972) and Schwartz et al. (1997). This ensures that two vowels that are more distant in the perceptual space are less penalized by the dispersion constraint. In other words, this cost reflects the confusability of vowel pairs.

$$(2) \quad disp_{i,j} = \frac{1}{(d_{i,j})^2}$$

390 To relate the cost of a vowel contrast to its probability, we use the maximum entropy
 391 framework (MaxEnt; Hayes and Wilson 2008), a probabilistic framework for constraint-
 392 based linguistic models. In this framework, a linguistic form is evaluated with respect to
 393 the weighted sum of its constraint violations, known as its harmony (Smolensky & Legendre,
 394 2006). In the dispersion model, there is a single constraint at play, so no weight is necessary.
 395 Crucially, this constraint is a systemic constraint: it evaluates pairs of vowels rather than
 396 individual vowels. We define the harmony of a vowel pair V_i and V_j , $h_{i,j}$, as the opposite of
 397 its cost in terms of dispersion, $disp_{i,j}$. This transformation allows for greater reward to less
 398 confusable vowel pairs.

$$(3) \quad h_{i,j} = -disp_{i,j}$$

399 The harmony of a pair of vowels is then exponentiated to obtain its maxent value,
 400 $maxent_{i,j}$. The exponentiation has two effects: (i) it transforms negative harmony values
 401 into positive values—a necessary step to get to probabilities (probabilities are always posi-
 402 tive), and (ii) it amplifies differences among candidate forms.

$$(4) \quad maxent_{i,j} = e^{h_{i,j}}$$

403 Maxent values are used to calculate the probabilities of candidates. Probabilities are
 404 obtained by normalizing by the sum of the maxent values of all candidates. In the present
 405 application, the set of candidates is the set of vowel pairs in the language. The probability
 406 of a pair vowel V_i and V_j can then be calculated as follows:

$$(5) \quad prob_{i,j} = \frac{maxent_{i,j}}{\sum_{i,j} maxent_{i,j}}$$

407 A last step is necessary to get from the probability of vowel pairs to the probability of
 408 individual vowels. We use the marginalization method proposed by Storme (2026) to eval-
 409 uate systemic constraints in MaxEnt. Concretely, the probability of a vowel V_i is obtained
 410 by summing the probability of all vowel pairs that include V_i as a member. This approach

411 predicts that a vowel that is involved in more distinct vowel pairs should be more frequent
 412 in the lexicon.

$$(6) \quad prob_i = \sum_j prob_{i,j}$$

413 3.4.3. Focalization model

414 The focalization model includes a constraint favoring focal vowels, that is vowels with ad-
 415 jacent formants ($F1-F2$, $F2-F3$, and $F3-F4$) in close proximity. Concretely, individual vowels
 416 are penalized according to the inverse square of the distance between their adjacent for-
 417 mants, as shown in Equation 7. Each pair of adjacent formants ($F1-F2$, $F2-F3$, and $F3-F4$)
 418 contributes to the overall focalization cost through summation, as in Schwartz et al. (1997).

$$(7) \quad foc_i = - \left(\frac{1}{(F2_i - F1_i)^2} + \alpha \frac{1}{(F3_i - F2_i)^2} + \alpha \frac{1}{(F4_i - F3_i)^2} \right)$$

419 Following a suggestion by Schwartz et al. (1997, p. 265), we add a free parameter, λ ,
 420 constrained to the $[0, 1]$ interval using a Beta distribution, allowing convergence between
 421 lower formants ($F1-F2$) to play a greater role than between higher formants ($F2-F3$, $F3-F4$) in
 422 the overall focalization cost. This asymmetric weighting of lower vs higher formants in the
 423 definition of the focalization cost parallels the dispersion model. We also test two variants
 424 of this model: one where λ is universal and one where it is language-specific, here again
 425 parallelling the dispersion model.

426 The constraint cost is defined as a positive number in the case of dispersion (Equation 2)
 427 but as a negative number in the case of focalization (Equation 7), following again Schwartz
 428 et al. (1997). This is due to the fact that dispersion *penalizes* small (inter-vowel) distances
 429 whereas focalization *favors* small (inter-formant) distances.

430 As in the dispersion model, the harmony is defined as the opposite of the focalization
 431 cost, allowing vowels with a smaller focalization cost (i.e. with greater formant conver-
 432 gence) to correspond to higher harmony, and therefore to be ultimately more frequent.

$$(8) \quad h_i = -foc_i$$

433 Other components of the model are the same as in the Dispersion Theory model, except

that the marginalization method is no longer needed in this case, since the focalization constraint is not a systemic constraint: it evaluates vowels individually, and not collectively.

Table 2 recapitulates the five models that were tested and how they differ with respect to their parameters. The baseline model has no free parameter. The universal versions of Dispersion Theory and Focalization Theory have a single parameter λ whereas their language-specific versions have as many parameters as there are languages in the sample (i.e. 30). All parameters are drawn from a Beta distribution, constraining lower formants to play a larger role in perception than higher formants.

Model	Free parameter	Prior
Baseline	none	
Dispersion Theory (universal)	λ	$B(2, 2)$
Dispersion Theory (language-specific)	λ_i	$B(2, 2)$
Focalization Theory (universal)	λ	$B(2, 2)$
Focalization Theory (language-specific)	λ_i	$B(2, 2)$

Table 2. Model parameters

3.5. Statistical analysis

All models were implemented as Bayesian multinomial regressions, using the `rjags` package (Plummer, 2025) in R (R Core Team, 2026). Three MCMC chains were used with 100,000 samples and a thinning interval of 10 (which means that every 10th value in the chain was kept in the final MCMC sample while all other values were discarded). The first 5,000 samples of each chain were used for burn-in (which means they were also discarded).

We used two diagnostics to ensure model convergence. Convergence of the chains on the posterior distribution was assessed using the Gelman-Rubin statistics. A value larger than 1.1 was taken as indicating that chains have not converged adequately (Kruschke, 2015, p. 181). Furthermore, we aimed for an effective sample size superior to 10,000 (Kruschke, 2015, p. 184). Effective sample sizes for all parameters estimated in the phonetically based models were superior to 10,000, indicating that the MCMC samples were large enough for stable and accurate numerical estimates of the posterior distributions (Kruschke, 2015, p. 184).

For model comparison, the Deviance Information Criterion (DIC; Gelman et al. 2013, pp. 172–173) was used. The DIC balances model fit and model complexity, therefore helping prevent overfitting. A model with a smaller DIC is a better model of the data. We use

the DIC for model comparison because it is directly available in `rjags` and can be computed straightforwardly from the fitted models. Although alternative Bayesian model comparison measures, such as the Watanabe-Akaike Information Criterion (WAIC) and leave-one-out cross-validation (LOO), have been advocated as more robust predictive criteria (Vehtari et al., 2017), their implementation requires additional computation and post-processing in `rjags`. Given that our primary aim is to compare the relative fit of a limited set of theoretically motivated models rather than to optimize predictive performance, DIC provides a practical criterion for model comparison in this context. We also compare observed and predicted counts to further illustrate how the models differ.

4. Results

4.1. Descriptive results

In total, the 30 lexicons from the DoReCO sample contain 232,282 short oral monophthongal vowels. Figure 3 illustrates the distribution of these vowels across the 20 vowel qualities attested in the corpus, reflecting their occurrence counts across the corresponding lexicons.

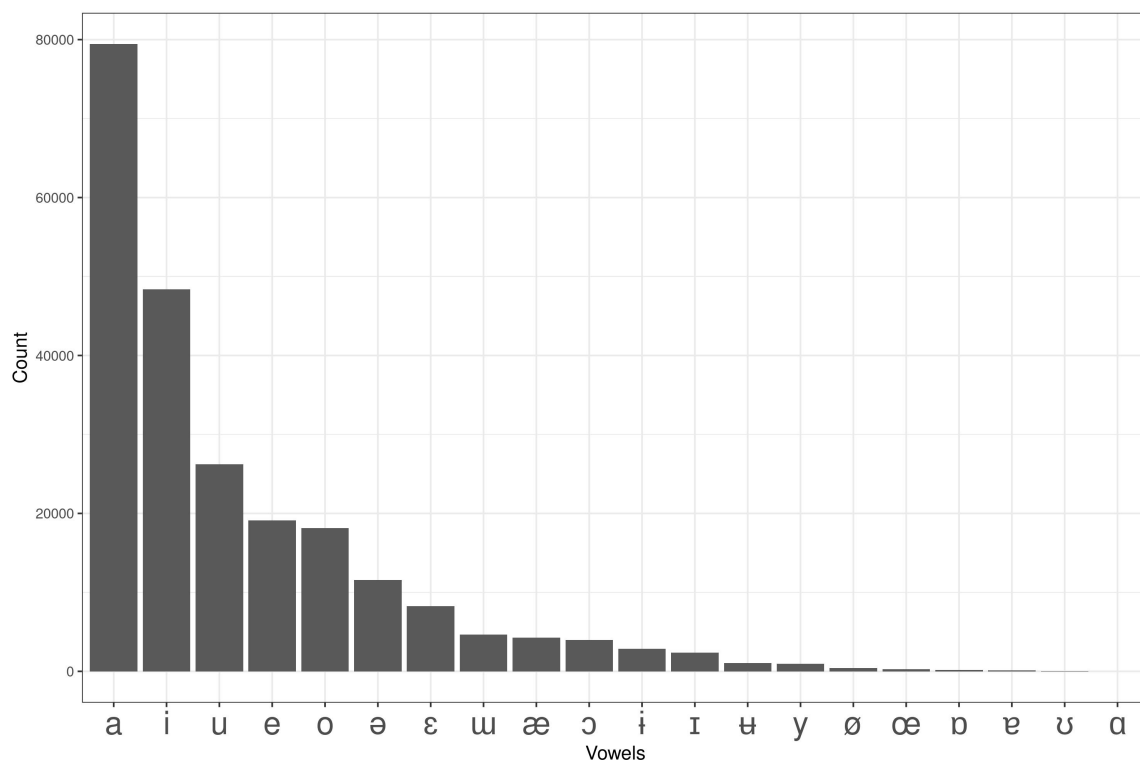


Figure 3. Vowel type count aggregated across the 30 lexicons

Type counts are strongly correlated with vowel counts across inventories, as confirmed by a Pearson correlation test ($r(18) = .866, p < .001, 95\%CI = [.687, .946]$). This is in line with Gordon (2016, p. 75). The distributions across inventories and across lexicons have different shapes though, as can be seen by comparing Figure 3 and Figure 2: counts decrease faster in lexicons (Figure 3) and slower in inventories (Figure 2). In inventories, we see a drop in after the 5th most frequent vowel ([e]) whereas the decrease is smoother in lexicons. This asymmetry is also reflected in Gordon (2016, p. 75). We return to this point in the discussion, where we provide a potential explanation.

Vowel counts in the corpora (token counts) and in lexicons (type counts) are extremely strongly correlated, $r(18) = .999, p < .001, 95\%CI = [.998, 1.000]$. This suggests that the same factors shape the distribution of vowels in lexicons and in corpora. We return to this point in the discussion. In practical terms, this allows us to focus on modeling type counts.

Figure 4 illustrates the average phonetic realization of short oral monophthongal vowels in each of the 30 languages in the sample. The values are plotted in Bark scale, with dimensions representing $F1$ and $F2'$, reflecting the auditory space used to calculate vowel contrast distinctiveness in the Dispersion Theory model. Formant values are consistent with vowel labels, suggesting that formant tracking was overall successful.

A note of caution is in order. In Bāinounk Gubëeher (Glottocode: bain1259), one vowel, [ɪ], is attested in the corpus but lacks a phonetic realization in our analysis. This is because [ɪ] appears only once in the entire corpus and not in the subset used to estimate phonetic realizations (see Section 3.3). Cobbinah (2013, pp. 167–168) notes that the phonemic status of [ɪ] is unclear due to the absence of clear minimal pairs. Given its single attestation in the corpus, its phonemic status is particularly uncertain. For this reason, and in the absence of phonetic data, we have excluded [ɪ] from the analysis. Consequently, for the purpose of this study, Bāinounk Gubëeher is treated as having 9 vowels rather than the 10 listed in Table 1. We return to the question of the phonemic status of vowels in the discussion.

4.2. Model comparison

Each model was fit to the count data in a single regression, but treating each language independently. For the phonetically based models (Dispersion Theory and Focalization Theory, with their universal and language-specific versions), phonetic realization of vowels in each

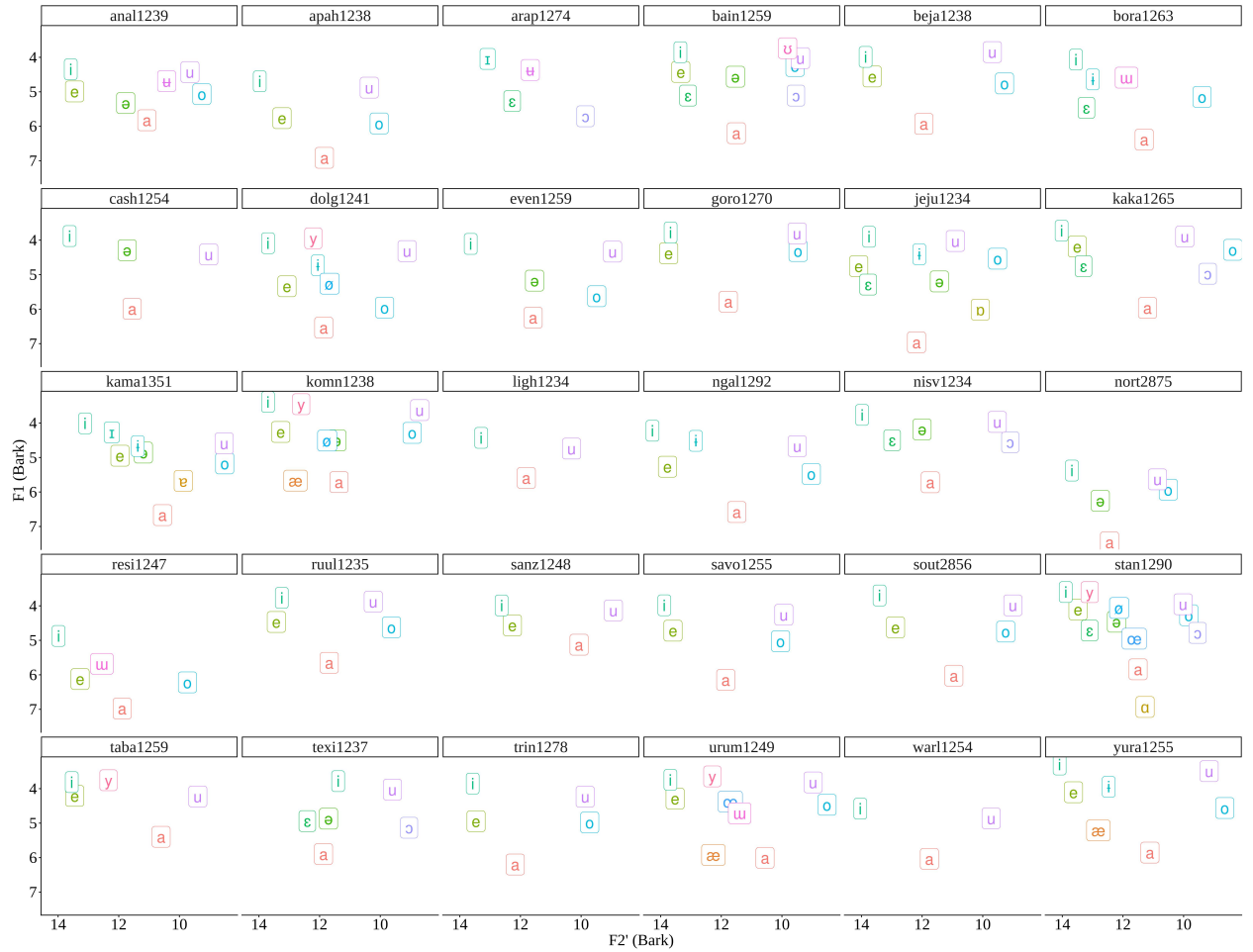


Figure 4. Vowel realization

language was used to predict vowel counts in the corresponding language.

Model comparison based on the Deviation Information Criterion (DIC) is presented in Table 3. The Dispersion Theory model with language-specific weighting of higher vs lower formants is by a very large margin the best model. Spiegelhalter et al. (2002) provide as a rule of a thumb that models with a difference in DIC between 3 and 7 have considerably less support. Here the difference in DIC with the second best model is 5072.27. This means that the data overwhelmingly support this model against the other models. A closer look at the deviance component of the DIC shows that the improvement is almost entirely driven by the model's goodness of fit, with its larger complexity being comparatively negligible. This large difference can be explained by the fact that vowel type counts are very large ($N = 232,282$), making the likelihood of the data under the best model much higher than under alternative models.

To get a better picture of goodness of fit, we plot the observed vs predicted counts for

Model	DIC	ΔDIC
Dispersion Theory (language-specific)	57196.93	0.00
Dispersion Theory (universal)	62269.20	5072.27
Focalization Theory (language-specific)	97730.76	40533.83
Focalization Theory (universal)	99518.75	42321.82
Baseline	100657.02	43460.10

Table 3. Model comparison

the five models in Figure 5. The figure visually confirms that dispersion models outperform both the baseline and focalization models, as their regression lines lie closer to the perfect prediction line ($y = x$).

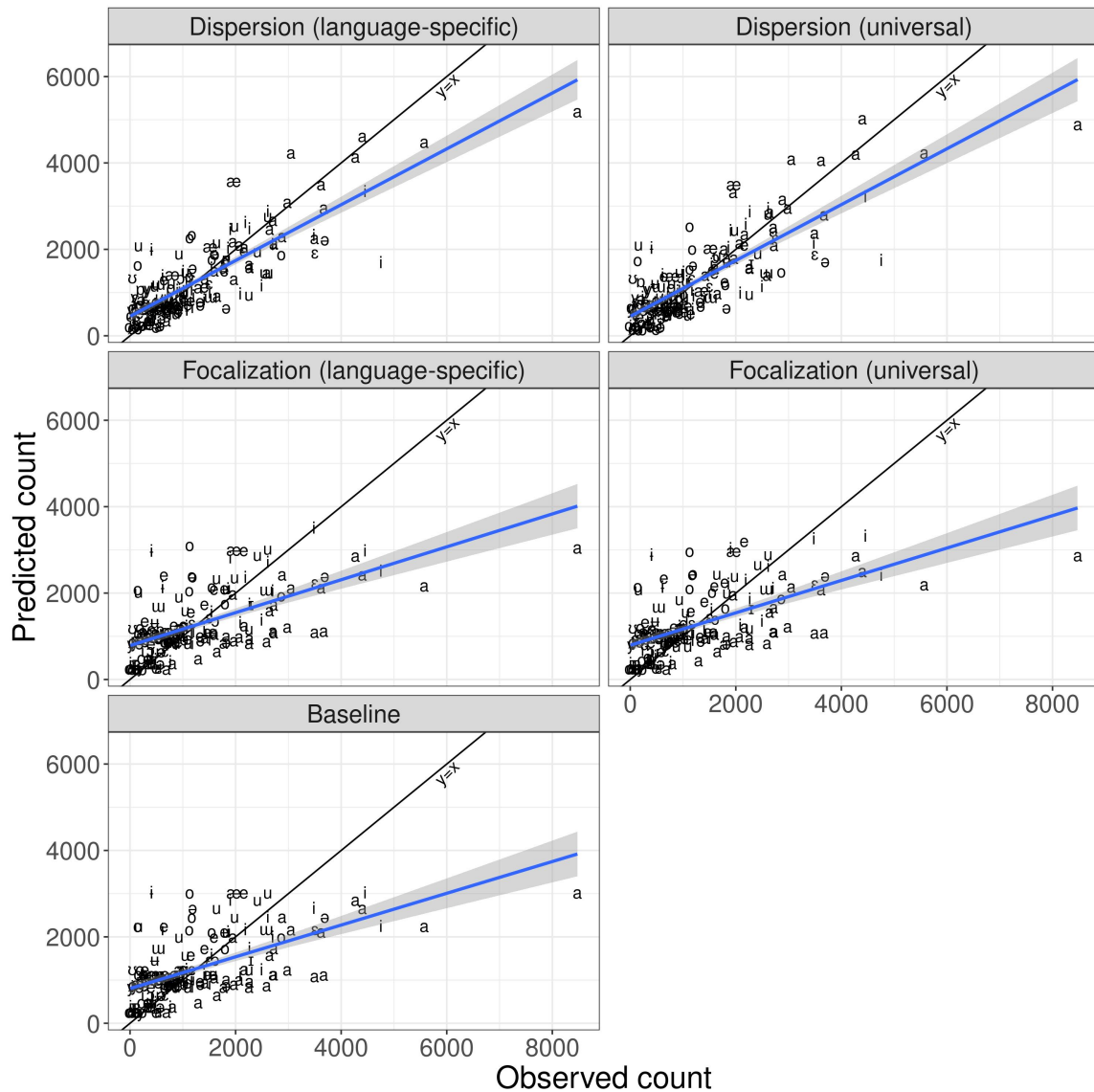


Figure 5. Observed counts vs counts predicted by the five models, with regression lines. The line $y = x$ corresponds to perfect prediction

Visual inspection of Figure 5 suggests that one key difference between dispersion and focalization models is their ability to predict the high lexical frequency of [a]. In focalization models, [a] is not overrepresented among vowels predicted to be highly frequent—all three corner vowels [a, i, u] are well represented, and so are some mid vowels (i.e., [e, o]) and central vowels (i.e., [ɪ]). In contrast, dispersion models predict that [a] dominates, aligning more closely with the patterns observed in lexicons. We will go back to this point in the discussion.

Figure 6 shows the predicted vowel counts by language under the best model, i.e. dispersion with language-specific weights. Inspection of the DIC at the level of each language reveals that the dispersion model outperforms the baseline and focalization models for 28 out of the 30 languages, confirming its superiority at the level of individual languages.



Figure 6. Observed vowel counts vs counts predicted by the best model, i.e. dispersion with language-specific weighting of higher vs lower formants

However the dispersion model performs worse with two languages: Evenki (Glottocode: even1259) and Swiss French (Glottocode: stan1290). In Evenki, schwa is the most frequent vowel in the lexicon; yet the dispersion model assigns it the lowest count (see Figure 6). This

follows from the fact that schwa is central in the auditory space (see Figure 4), and therefore spectrally close to the other vowels in the inventory. The difficulty to derive the behavior of schwa in Dispersion Theory is well known (see Section 2.1). Modeling its behavior would require a richer model that likely includes an effort minimization constraint. In the case of Evenki, this is made particularly plausible by the fact that schwa is on average the shortest vowel in the corpus (53 ms).

Another factor that may increase the type frequency of [ə] in Evenki is vowel harmony. Evenki has a vowel harmony process whereby [a] alternates with [ə] in suffixes depending on the stem vowel, e.g. [urə-βə] ‘mountain-ACC’ - [bira-βa] ‘river-ACC’ (Bulatova & Grenoble, 1999, p. 4). Some surface schwas may therefore correspond to an underlying phoneme /a/. This reveals one limitation of the approach focusing on transcriptions, without controlling for alternations. We go back to this limitation in the discussion.

Swiss French is the second language for which the dispersion model actually provides a worse theory than alternative models. Swiss French is the language with the largest inventory size (N=12). The fact that Dispersion Theory does not work very well for larger inventories is not new (see Schwartz et al. 1997). But there is a specific problematic prediction that deserves further attention. In Swiss French, [ɑ] is the least frequent vowel in the lexicon extracted from DoReCo (it is attested in only 5 distinct words in the corpus); yet the dispersion model assigns it the highest count (see Figure 6). This follows from the fact that [ɑ] has the highest *F1* value of all vowels in the corpus (see Figure 4) and therefore is very distinct from other vowels along this dimension.

As for Evenki, this problematic prediction might also be partly due the neglected role of vowel duration. [a] and [ɑ] differ temporally in Swiss French, with [ɑ] being longer than [a] (e.g., *pâte* [pat] ‘dough’ - *patte* [pat] ‘leg’; Racine and Andreassen 2012, pp. 185–186). Dispersion Theory does not take duration into account and therefore might fail with languages where vowel quality distinctions are enhanced by temporal cues. We return to this point in the discussion.

Duration might also explain the unexpectedly high *F1* value of Swiss French [ɑ] in DoReCO. In Swiss French, [ɑ] is actually reported to have a *lower F1* realization than [a] (Racine & Andreassen, 2012, pp. 185–186), contra our results. This discrepancy is likely due to a difference in duration patterns of [ɑ] in the two studies. In DoReCo, [ɑ] is more

563 than twice longer (193 ms on average) than [a] (91 ms on average), whereas the difference
 564 in Racine and Andreassen (2012) is much smaller (59 ms vs. 44 ms). This means that Swiss
 565 French [a] is likely hyperarticulated in DoReCO and therefore undergoes less target under-
 566 shoot (Lindblom, 1963) than other vowels, explaining its high $F1$ realization and therefore
 567 its higher distinctiveness along $F1$.

568 Figure 7 displays the 30 language-specific λ parameters, which weight the contribution
 569 of $F2'$ relative to $F1$ in the perception of vowel contrasts in the best dispersion model.

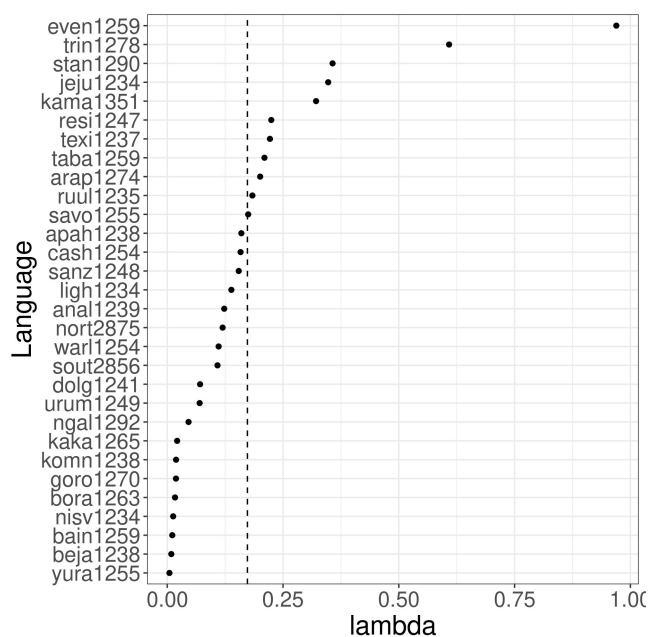


Figure 7. Language-specific λ parameters weighting $F2'$ in the dispersion model. The average value for this parameter across languages is represented by the dashed line

570 The average λ value across languages is smaller than 1, meaning that the model requires
 571 $F2'$ to have a smaller perceptual weight than $F1$, in line with previous results on vowel
 572 inventories (Schwartz et al., 1997). However, language-specific variation in λ improves the
 573 model's fit, as shown by the lower DIC of the model with language-specific weights (see
 574 Table 3).

575 The high λ value for Evenki should be interpreted cautiously, as the model does not
 576 perform well for this language. In some languages, λ is close to zero, indicating that only
 577 dispersion along $F1$ matters for predicting the lexical distribution of vowels. We discuss a
 578 potential explanation for the dominance of $F1$ in the discussion.

5. Discussion and conclusion

The data provide evidence for the role of perceptual distinctiveness in the lexical distribution of vowels, in line with earlier claims (Gordon, 2016) and with predictions from speaker- and listener-based models of sound change. Section 5.1 examines the implications of these results for comparing dispersion and focalization as models of typological tendencies in vowel systems. Section 5.2 explores explanations for the dominance of $F1$ over $F2'$. Section 5.3 addresses the very strong correlation between vowel counts in lexicons and in corpora, discussing its potential implications for whether effects of perceptual distinctiveness arise in word formation or word usage. Section 5.4 highlights the limitations of this work and outlines future directions.

5.1. Dispersion vs focalization

An interesting implication of the results is that dispersion is a far stronger predictor of vowel lexical frequency than focalization. This is notable because the two theories are sometimes treated as equally plausible explanations for vowel inventory structure (e.g., Johnson 2012, pp. 141–142). For example, both theories account for the high frequency of [a, i, u] across languages. Dispersion Theory attributes this to their status as corner vowels in the perceptual space, while Focalization Theory explains it in terms of formant convergence ($F1$ – $F2$ for [a] and [u], and $F3$ – $F4$ for [i]).

However, the predictions of the two theories diverge more strongly when examining the frequency distribution of vowels in lexicons. In this case, we observe greater separation among [a], [i], and [u] across languages (see Figure 3), as well as within languages (see Figure 6), with [a] being often more frequent than [i], which in turn is often more frequent than [u]. Dispersion Theory outperforms Focalization Theory in predicting this asymmetry.

There is a clear explanation for this: [a] typically has the highest $F1$ value in a vowel system and is often the only vowel with that value. In contrast, [i] and [u] share similarly low $F1$ values, as both are high vowels. If $F1$ plays a significant role in vowel distinctiveness, it follows that [a] will be particularly distinct from other vowels. By contrast, focalization favors vowels with close formant proximity, but this metric does not specifically advantage [a]. For example, [u] also exhibits $F1$ – $F2$ convergence due to its very low $F2$. Even if $F1$ –

F2 convergence is weighted more heavily—reflecting the greater perceptual importance of lower formants—this would still predict [u] to be quite frequent. As a result, focalization fails to capture the dominance of [a] observed in the data (see Figure 5).

Dispersion Theory also explains why [i] tends to rank second to [a] in lexical frequency. According to De Boer (2011), [i] generally has a more extreme (lower) *F1* value than [u], a pattern confirmed in most languages in our corpus (see Figure 4). This results in greater *F1* dispersion for [i] than for [u], accounting for its higher frequency. Notably, Focalization Theory combined with a greater weighting of *F1–F2* convergence does not predict this asymmetry, as [i]’s *F1* and *F2* values are widely separated.

We propose the following explanation for why Dispersion Theory appears to be a stronger predictor than Focalization Theory for lexical frequency patterns compared to vowel inventories: Sound inventories are subject to a specific pressure to maximize the number of phonological contrasts (Flemming, 2004). In inventories, this pressure boosts the frequency of vowels that are not as perceptually distinct from other vowels as [a] and which happen to be also characterized by formant convergence (e.g., [u]). However, when considering lexicons, this inventory-level pressure no longer applies, allowing the effects of perceptual distinctiveness to emerge more clearly. The advantages of [a] in terms of *F1* distinctiveness then become more evident. The hypothesis that there is an inventory-specific pressure to maximize the number of vowel contrasts also explains why the frequency of sounds declines more gradually in inventories than in lexicons (see Section 4.1).

5.2. Explaining the dominance of *F1*

For the lexical distribution of vowels to emerge, *F1* must play a far greater role in vowel perception than *F2*. This is in line with what has been found in vowel inventories, where the greater weighting of *F1* explains why there are typically more vowel contrasts along the height dimension than along the front-back dimension (Schwartz et al., 1997). It also explains why there are so-called ‘vertical’ vowel inventories, exploiting only the height dimension for vowel contrasts (e.g., Marshallese, Kabardian, etc.; see Flemming 2004, pp. 250–251), but no reported ‘horizontal’ vowel inventories, exploiting only the front-back dimension.

This dominance of *F1* may stem from its greater importance in perception under noisy conditions (see Schwartz et al. 1997, p. 276). Listeners have been shown to rely more on

F1 than *F2* to identify vowels in noise (Parikh & Loizou, 2005), and the role of *F1* as a cue to phonological contrasts increases under such conditions (Carranante et al., 2024).

The prominence of *F1* suggests that the lexical distribution of vowels is optimized for worst-case scenarios, i.e., listening in noisy environments. In listener-based models of sound change (e.g., Blevins 2004; Ohala 1981), this would imply that noisy conditions generally dominate, gradually leading to a decline of vowels that are less robust to noise. In speaker-based models (e.g., Flemming 2017; Lindblom 1990; Martin 2007), it would mean that speakers optimize word forms for worst-case scenarios, favoring sounds that remain perceptible even in noise.

5.3. Explaining the very high correlation between type and token counts

We found that lexical and corpus counts of vowels are almost perfectly correlated in our 30-language sample ($r = .999$; see Section 4.1). This indicates that the relationship between lexical and corpus counts is nearly linear. Such a linear relationship would not be expected if word usage were optimized for perception beyond what is already reflected in the lexicon. Specifically, if speakers preferentially selected words with more distinct vowels when producing a sentence, we would expect dispersed vowels like [a] to receive an additional frequency boost beyond their lexical representation, resulting in a non-linear relationship between lexical and corpus frequency.

In other words, dispersion likely influences lexical creation but not word usage. This distinction can be interpreted differently depending on the model of sound change. In a speaker-based model (e.g., Martin 2007), speakers may apply dispersion principles when creating new words, favoring phonological forms with highly distinct vowels (e.g., preferring forms containing [a] over those with less distinct vowels). However, during communication, they do not select lexical forms based on such properties; enhancement of phonetic properties in the presence of noise (e.g., Lombard effect) would only happen after lexical selection. In a listener-based model (e.g., Ohala 1981), speakers may misperceive words with confusable vowels, but when speaking, they select words independently of vowel dispersion.

5.4. Limitations and future directions

This study has several limitations that could be addressed in future research. First, our analysis relies on DoReCo’s transcriptions and word segmentation, which may be problematic in languages with phonological alternations, such as the [a]-[ə] alternations in Evenki. Since Dispersion Theory focuses on phonological contrasts, an ideal approach would compare phonemes rather than surface forms. Additionally, we adopt DoReCo’s definitions of long and short vowels, but these may lack accuracy, particularly in languages where spectral contrasts interact with temporal cues. For example, the [a]-[ɑ] contrast in Swiss French—a challenging case for Dispersion Theory—relies on both temporal and spectral cues. It also requires careful control for target undershoot, as durational differences can influence formant values.

We should also better account for phonotactics. For vowels, stress is particularly relevant due to vowel reduction in unstressed syllables, which affects both vowel realization and perceptual distinctiveness (Flemming, 2004). Furthermore, morphological factors should be controlled for, as the frequency of certain vowels in some languages may stem from morphemes containing those vowels. However, controlling for these factors would require a corpus with comprehensive phonological and morphological annotations (e.g., stress placement, morpheme boundaries).

The best model retains some limitations of Dispersion Theory, particularly its difficulty in modeling the behavior of schwa, which has been argued to be neutral with respect to other vowels in dispersion. A richer model that accounts for vowel duration and phonotactics would be necessary to address this. Schwartz et al. (1997) argues that the best model of vowel inventories combines both dispersion and focalization. In this paper, we have only reported comparisons between simple models and not their combination. We attempted to combine both, but the models failed to converge, likely due to the increased number of parameters to estimate, including the λ parameters weighting lower and higher formants differently and a weight distinguishing the contributions of dispersion vs. focalization.

Finally, another limitation concerns how to explain the weights. Here, we adopted a data-driven approach, identifying the best weights to fit vowel counts in the lexicons. However, one would ultimately want to determine whether these weights can be explained by independent properties of the language. We attempted to correlate them with inventory

696 size, height distinctions, and backness distinctions (see Supplementary files), but no clear
697 relationship emerged. For now, it remains an open question why languages differ in how
698 they weigh cues. This could be related to other dimensions not considered here, such as the
699 full system of phonological contrasts (including long vowels, consonants, stress, etc.).

Language	Glottocode	Family	Area	Word tokens	Reference
Anai	anal1239	Sino-Tibetan	Eurasia	13015	Ozerov (2024)
Yali (Apahapsili)	apah1238	Nuclear Trans New Guinea	Papunesia	7474	Riesberg (2024)
Arapaho	arap1274	Algic	North America	4746	Cowell (2024)
Bainounk Gubéeher	bain1259	Atlantic-Congo	Africa	11477	Cobbinah (2024)
Beja	beja1238	Afro-Asiatic	Africa	15439	Vanhove (2024)
Bora	boral263	Boran	South America	8431	Seifart (2024a)
Cashinahua	cash1254	Pano-Tacanan	South America	10043	Reiter (2024)
Dolgan	dolg1241	Turkic	Eurasia	8778	Däbritz et al. (2024)
Evenki	even1259	Tungusic	Eurasia	8315	Kazakevich and Klyachko (2024)
Gorwaa	goro1270	Afro-Asiatic	Africa	10674	Harvey (2024)
Jejuan	jeju1234	Koreanic	Eurasia	7272	Kim (2024)
Kakabe	kaka1265	Mande	Africa	9525	Vydrina (2024)
Kamas	kama1351	Uralic	Eurasia	10052	Gusev et al. (2024)
Komnzo	komn1238	Yam	Papunesia	9892	Döhler (2024)
Light Warlpiri	ligh1234	Mixed Language	Australia	8672	O'Shannessy (2024a)
Dalabon	ngal1292	Gunwinyguan	Australia	3412	Ponsonnet (2024)
Nisvai	nisv1234	Austronesian	Papunesia	11872	Aznar (2024)
Northern Alta	nort2875	Austronesian	Papunesia	8544	Garcia-Laguia (2024)
Resígaro	resi1247	Arawakan	South America	7985	Seifart (2024b)
Ruuli	ruul1235	Atlantic-Congo	Africa	8718	Witzlack-Makarevich et al. (2024)
Sanzhi Dargwa	sanz1248	Nakh-Daghestanian	Eurasia	5146	Forker and Schiborr (2024)
Savosavo	savo1255	Isolate	Papunesia	12075	Wegener (2024)
Nafsan (South Efate)	sout2856	Austronesian	Papunesia	10243	Thieberger (2024)
French (Swiss)	stan1290	Indo-European	Eurasia	13339	Avanzi et al. (2024)
Tabasaran	taba1259	Nakh-Daghestanian	Eurasia	5059	Bogomolova et al. (2024)
Texistepec Popoluca	texi1237	Mixe-Zoque	North America	8473	Wichmann (2024)
Mojeño Trinitario	trin1278	Arawakan	South America	7792	Rose (2024)
Urum	urum1249	Turkic	Eurasia	10035	Skopeteas et al. (2024)
Warlpiri	warl1254	Pama-Nyungan	Australia	7171	O'Shannessy (2024b)
Yurakaré	yura1255	Isolate	South America	20206	Gipper and Ballivián Torrico (2024)

Table 4. 30-language sample: metadata

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